

It must be modified depending on the type of analysis:

0: sample from prior

1: clock model selection

2: approximated likelihood (gradient and Hessian)

3: divergence time estimation

Evolutionary
model

```
seed = -1          * Seed number, type -1 to generate a random seed number
seqfile = ../concatenated.phylip * Path to input sequence file (e.g., many alignments in a unique file)
treefile = ../startingTree.phy   * Path to input tree file including calibration points as node labels
outfile = out.log   * Path to log output file
mcmcfile = mcmc.txt * Path to output file with MCMC samples

ndata = 5          * Number of partitions
seqtype = 0        * 0: nucleotides; 1:codons; 2:AAAs
usedata = 1        * 0: no data (samp from prior); 1:seq likelihood; 2:approx likelihood (use in.BV); 3:out.BV
clock = 1          * 1: global clock; 2: independent rates; 3: correlated rates

model = 4          * 0:JC69, 1:K80, 2:F81, 3:F84, 4:HKY85. Pick 4 for nt, and 2 for aa
alpha = 0.5        * alpha for gamma rates at sites
ncatG = 5          * No. categories in discrete gamma

cleandata = 0      * remove sites with ambiguity data (1:yes, 0:no)?
```

priors
birth-death
process

```
BDparas = 1 1 0.1 * birth, death, sampling
kappa_gamma = 2 .2 * gamma prior for kappa
alpha_gamma = 2 4  * gamma prior for alpha
rgene_gamma = 2 20 * gamma prior for overall rates for genes
sigma2_gamma = 1 10 * gamma prior for sigma^2 (for clock=2 or 3)
```

MCMC
sampling

```
print = 1          * avoid printing branch rates for now as the tree is very large
checkpoint = 1      * turn checkpointing on with 1, change to 2 if resume is required after a failed one where 1
burnin = 4000       * Number of iterations that will be discarded as part of burn-in
sampfreq = 6        * Sampling frequency
nsample = 20000      * Number of samples to be collected
```

* burnin, samplefreq, and nsample imply that a total burnin + (sampfreq * nsample) samples will be collected with mcmc